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Brief report

Visuomotor strategies for collision avoidance in athletes: Insights from virtual reality

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ABSTRACT

This study explores visuomotor control in athletes for collision avoidance using virtual reality. Thirty-nine athletes navigated dynamic scenarios, pursuing a virtual target while avoiding two to five virtual defenders. Confirmatory Factor Analysis validated a model that captured two features of scanning behavior based on head movements recorded during activity: the overall amount and temporal pattern. Linear mixed models showed that these features significantly differentiated successful from unsuccessful defender avoidance ($p < .05$), suggesting that efficient environmental scanning is crucial for collision avoidance while highlighting the potential of visuomotor interventions to reduce collision-related sport injuries.
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1. Introduction

The highest injury rates in contact- and collision-based sports typically stem from falls and collisions with other athletes.[1,2] Unanticipated collisions often lead to head accelerations that may cause concussions[3] and musculoskeletal injuries.[4] Understanding visuomotor control for effective collision avoidance is thus crucial for developing behavioral interventions to mitigate these injuries.

Visuomotor control involves using visual information to regulate action. In accordance with ecological theory, this requires athletes to actively seek visual information as action unfolds.[5] Effective navigation (i.e., their direction of travel, or heading) and collision avoidance in sports necessitates athletes coordinating their eyes and head movements to efficiently scan their environment and dynamically relate to obstacles.[6,7] This is particularly critical for avoiding collisions with opponents attempting to intercept the athlete's path surreptitiously.[8]

Prior research shows that analyzing eye movements provides insights into visuomotor control for collision avoidance, with less efficient eye movements on computer-based tracking tasks linked to higher collision risk during competition.[9] However, previous studies did not assess how visuomotor behavior unfolds, in real time, as athletes navigate against defenders in contact sports.

We leveraged advancements in virtual reality (VR) technology[10–12] to record visuomotor activity during a dynamic, overground sport-like navigation task requiring collision avoidance of virtual defenders during moving target pursuit. Our aim was to identify visual scanning behaviors related to collision outcomes in these human–virtual defender interactions. We hypothesized that the amount and pattern of change in athletes' head movements—proxies supportive of scanning behavior[13]—would differentiate interactions leading to virtual collisions from those that did not.

2. Method

Thirty-nine healthy athletes (age = 20.40 ± 2.57 years; 14 female) from sanctioned competitive collegiate club or varsity contact sport teams (Median 4.5 ± 5.8 years experience), provided informed consent or assent with parental/guardian consent. Participants ran during a virtual athletic navigation task in a 14 × 13 m gym visualized as a stadium, using an Oculus Quest 2 HMD (6 DoF head tracking; 1832 × 1920 resolution per eye at 72 Hz; Meta Platforms, Inc., Menlo Park, CA). The task involved sprinting to intercept a moving virtual humanoid target while avoiding two to five adaptive virtual defenders, positioned at least 1.5 m away from the human to block the interception path. The

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number and layout of defenders varied but remained consistent across conditions (see Fig. 1; Kiefer et al.[14]). Participants underwent 48 trials across two 24-trial blocks, separated by a 5-minute rest, including both left and right starting positions. Prior to the trials, participants acclimated through walking and jogging in VR and completed three speed trials to standardize target speed and maximum defender speed at 70 % of the participant's maximum sprint speed.

Head rotation data (x, y, z axes corresponding to up-and-down, side-to-side rotation, side-to-side lateral flexion movements, respectively) were sampled at 72 Hz. Human/virtual human interactions were defined as the period from when a participant entered a 1.5 m radial boundary of a defender until the first data frame when the human position was beyond the virtual defender in the forward (z) axis passing them. Interactions lasting 10 or more data frames (i.e., 150 ms) were analyzed.

We used Confirmatory Factor Analysis (CFA) to validate a measurement model of scanning behavior based on characteristics of the recorded 3D head rotations. CFA is a widely accepted approach to create measurement models from multifactor indicators.[15] The model was defined with two factors (F1 and F2). F1 reflects the overall amount of space covered by the scanning behavior. F2 captures the temporal structure of the scanning behavior. Variance in head angular position (Var_x, Var_y, Var_z) was used in Factor 1 indicators. Fluctuation magnitude in head position (Flu_x, Flu_y, Flu_z)—see computation details below—was used in Factor 2 indicators. CFA was conducted using the lavaan package in R (ver. 4.3.2; 2023-10-31) to validate the measurement model.

Fluctuation magnitude was computed using Fluctuation Intensity Analysis,[16] applied to time series of head angular position in each athlete–defender interaction. Time series were segmented at boundaries (k) defined by sign changes indexing head movement direction changes. The absolute difference between the head angular position at consecutive "k" boundaries was divided by the number of data points between the boundaries, producing a rate of change metric reflecting head movement speed in a direction before direction changes (i.e., scanning behavior). A fluctuation index, summing each segment's rate of change, measured fluctuation magnitude, where higher values indicate quicker and more frequent changes in head direction changes. This method, validated by prior work, correlates well with standard measures like sample entropy for analyzing short, non-stationary time series.[16]

3. Results

Across defender conditions the count of non-collision events consistently exceeded those of collisions, with non-collision counts ranging from 223 to 347 (Total: 1237; M = 309.25; SD = 57.97), and collision counts ranging from 21 to 30 (Total: 108; M = 27; SD = 4.24).

3.1. Measurement model of scanning behavior

CFA was performed on data obtained from 1245 interactions grouped within 38 clusters identified by participant ID. The CFA model

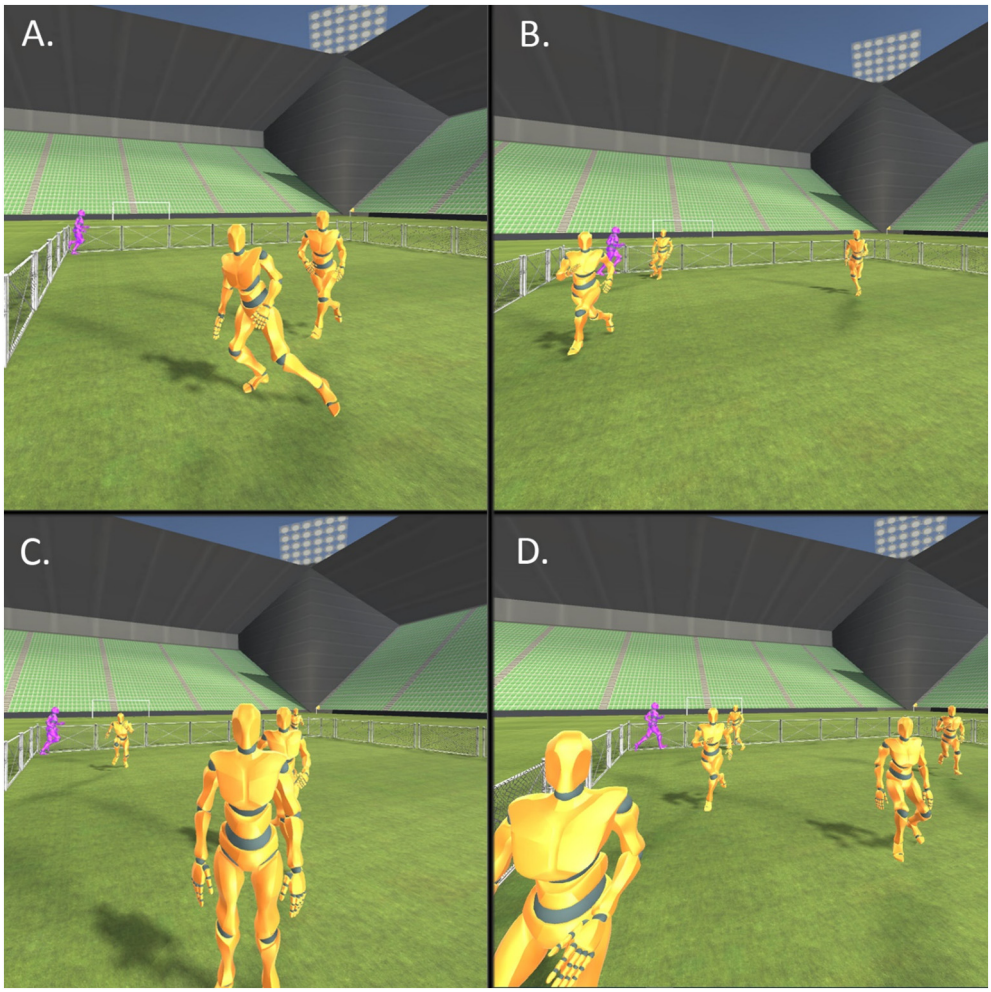

Fig. 1. First-person view of the VR scenario at the beginning of the trial, with initial positions of the moving target (purple humanoid figure) and defenders (orange humanoid figures) at the start of each trial for each of the four conditions: A) 2 virtual defenders, B) 3 virtual defenders, C) 4 virtual defenders, and D) 5 virtual defenders. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

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was estimated with adjusted standard errors that are robust to the clustering effect associated with repeated measures.

Standardized factor loadings for each indicator variable on their respective factor are displayed in Table 1. The moderate-to-strong and significant loadings (all $p < .001$) along with the adequate indices of robust fit (Comparative Fit Index = 0.976; Tucker-Lewis Index = 0.954; RMSEA = 0.085) supports the hypothesized structure of the measurement model. Thus, scores of F1 and F2 were used to characterize scanning behavior.

3.2. Scanning behavior & collision avoidance

Two mixed effect models were estimated to determine whether the amount and structure of scanning behavior—characterized, respectively, by the scores of F1 and F2 in our measurement model—are related to collision occurrence during our navigation task after controlling by task condition.

Model 1 showed a significant relationship between collision occurrence and the amount of scanning behavior, as indexed by F1 scores, $F(1, 1311) = 4.23, p = .04$. This relationship was qualified by an interaction between F1 score and condition, $F(3, 1310) = 4.49, p = .004$. Follow-up analyses showed that participants' F1 score was significantly higher—indexing greater overall amount of head rotations—in interactions that led to collision avoidance, $p = .001$, but only in the condition with 2 virtual opponents.

Model 2 showed a significant relationship between collision occurrence and F2 score, $F(1, 1312) = 6.03, p = .01$, which was not moderated by a significant Collision × Condition interaction. Regardless of condition, participants' F2 score was significantly higher (indexing a more richly varied structure of head movements) during interactions that led to collision.

4. Discussion

The present study examined visuomotor behavior by analyzing head movements within the context of collision avoidance in a sport-like VR scenario. This approach enabled us to identify, for the first time, a direct correlation between athletes' success in collision avoidance and their visual scanning behavior, characterized by specific patterns in the quantity and temporal structure of head movements. Specifically, CFA validated a measurement model of scanning behavior based on recorded head movements. The models distinguished two features of scanning: the overall amount and temporal pattern of head movements. Linear mixed models further highlighted the significance of this measurement model, showing a clear distinction between successful and unsuccessful avoidance interactions based on the identified features. This integration of visuomotor control with practical avoidance behaviors underscores the potential importance of targeted visuomotor strategies for athlete safety. These results also indicate that VR may be a powerful tool in developing interventions to train modifiable visuomotor efficiency,[17,18] ultimately reducing the risk of sport-related injuries.

While the VR scenarios allow for the controlled study, they do introduce some inherent limitations. For example, real-world player-to-player interactions typically involve a more complex set of cognitive and emotional factors that could serve to change how visuomotor strategies evolve over time and in the face of changing pressure contexts.[19,20] While our protocol presented conditions with a greater number of virtual defenders (i.e., increased perceptual loading), additional work focused on additional cognitive and emotional pressures, including fear of collision, would likely better replicate the conditions of competitive sport. In addition, while head rotation data can serve as a proxy for visual scanning strategies, the absence of eye tracking data makes it difficult to draw definitive conclusions around the specific targets of participants' perceptual activity during the various interactions captured in the current study, while a lack of locomotor biomechanics limits action assessment. With new technologies emerging that allow for embedded eye tracking within an HMD (cf. Meta Quest Pro, for example), future research should further explore this dynamic eye–head relationship.

Table 1
Standardized factor loadings for scanning behavior measurement model.

Indicator	Amount of scanning (F1)	Structure of scanning (F2)
Varx	0.552	–
Vary	0.601	–
Varz	0.576	–
Flux	–	0.835
Fluy	–	0.914
Fluz	–	0.838

5. Conclusion

The current research emphasizes the critical role visual scanning behavior that plays in collision avoidance, offering important insights for enhancing athlete safety through new training programs. Future studies should look to enhance the scanning amount and fluctuations to determine whether those changes lead to reduced collisions. The results indicate VR technology's potential to enhance understanding of visuomotor demands in sport; however, future research is needed to confirm if VR-based interventions can effectively reduce collision-related injuries.

CRediT authorship contribution statement

Adam Kiefer – Led the conceptualization, methodology, and original draft writing. Was also the principle investigator for funding acquisition and supervision/mentorship.

Nikki Aitcheson-Huehn – Co-led the investigation, data curation and project administration. Assisted in review and editing of writing.

Cortney N. Armitano-Lago – Co-led the investigation, data curation and project administration. Also assisted in the methodology development and the validation of core technologies. Finally, assisted in review and editing of writing.

Ryan MacPherson – Participated in the conceptualization of the project. Led methodology and core software development, as well as the data curation. Assisted in the review and editing of writing, as well as with the visualization of the figure.

Dominic Willoughby – assisted with the investigation, data curation and project administration. Assisted in review and editing of writing.

Paula L. Silva – Co-led the conceptualization, methodology and original draft writing. Was the co-investigator for funding acquisition and helped with supervision and mentorship on the project.

Declaration of generative AI and AI-assisted technologies in the writing process statement

During the preparation of this work the author(s) used ChatGPT4 in order to improve conciseness in the existing human-generated writing in adherence of the word count requirements. After using this tool/service, the author(s) reviewed and edited the content as needed and take full responsibility for the content of the publication.

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Confirmation of ethical compliance

This research was approved as protocol #19-2176 by the Institutional Review Board at the University of North Carolina at Chapel Hill.

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Declaration of interest statement

None.

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